Turn in 7

Grant Talbert
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Boston University

Problem 1

Clearly, $\alpha^2 \in F(\alpha)$. Since $F(\alpha)$ has odd degree over F, it necessarily has finite degree over F, and thus $F(\alpha)$ is a finite extension of F. Since every finite extension is algebriac, $F(\alpha)$ is algebraic, and thus all elements of $F(\alpha)$ are algebraic over F. Since $\alpha^2 \in F(\alpha)$, it's algebraic over F.

Since $\alpha^2 \in F(\alpha)$, clearly $F(\alpha^2) \subseteq F(\alpha)$. For the converse, consider the polynomial $x^2 - \alpha^2 \in F(\alpha^2)$. Clearly this splits into $(x-\alpha)(x+\alpha)$. Suppose for contradiction that $\alpha \notin F(\alpha^2)$. Then this polynomial does *not* split over $F(\alpha^2)$, and is thus clearly irreducible. It follows that $x^2 - \alpha^2$ is the minimal polynomial of α in $F(\alpha^2)$, and thus it's splitting field $F(\alpha, \alpha^2)$ over $F(\alpha)$ has

$$\left[F(\alpha, \alpha^2) : F(\alpha^2)\right] = \deg(x^2 - \alpha^2) = 2.$$

However, since $\alpha \in F(\alpha)$, we have $F(\alpha, \alpha^2) = F(\alpha)$, and thus $\left[F(\alpha), F(\alpha^2)\right] = 2$.

Now note that

$$[F(\alpha) : F] = \left[F(\alpha) : F(\alpha^2)\right]\left[F(\alpha^2) : F\right] = 2\left[F(\alpha^2) : F\right].$$

Since this is clearly an even number and not an odd number, this is a contradiction, and thus $\alpha \in F(\alpha^2)$. Therefore, we easily obtain $F(\alpha) \subseteq F(\alpha^2)$, and the fields are equal.

Problem 2

Let $x = \sqrt{3 - \sqrt{6}}$. Then $-x^2 + 3 = \sqrt{6}$, and thus $x^4 - 6x^2 + 9 = 6$. I claim the minimal polynomial is thus $p(x) = x^4 - 6x^2 + 3$. To verify irreducibility, note that clearly no roots of p exist in $\mathbb{Q}$. Extensive algebra that I do not feel like typing up shows that no such quadratic factorization exists either.

Solving for the roots of $p(x)$ gives

$$\begin{aligned}
& x^4 - 6x^2 + 3 = 0 \\
\implies & x^4 - 6x^2 + 9 = 6 \\
\implies & \left(-x^2 + 3\right)^2 = 6 \\
\implies & -x^2 + 3 = \pm\sqrt{6} \\
\implies & x^2 = 3 \pm \sqrt{6} \\
\implies & x = \pm\sqrt{3 \pm \sqrt{6}}.
\end{aligned}$$

We thus have roots

$$\left\{\sqrt{3+\sqrt{6}}, -\sqrt{3+\sqrt{6}}, \sqrt{3-\sqrt{6}}, -\sqrt{3-\sqrt{6}}\right\}.$$

Clearly you cannot construct $\sqrt{3+\sqrt{6}}$ using the elements of $\mathbb{Q}(\alpha)$, and so $p(x)$ does not split in $\mathbb{Q}(\alpha)$.